

Activities on Grid Technology at GISTDA

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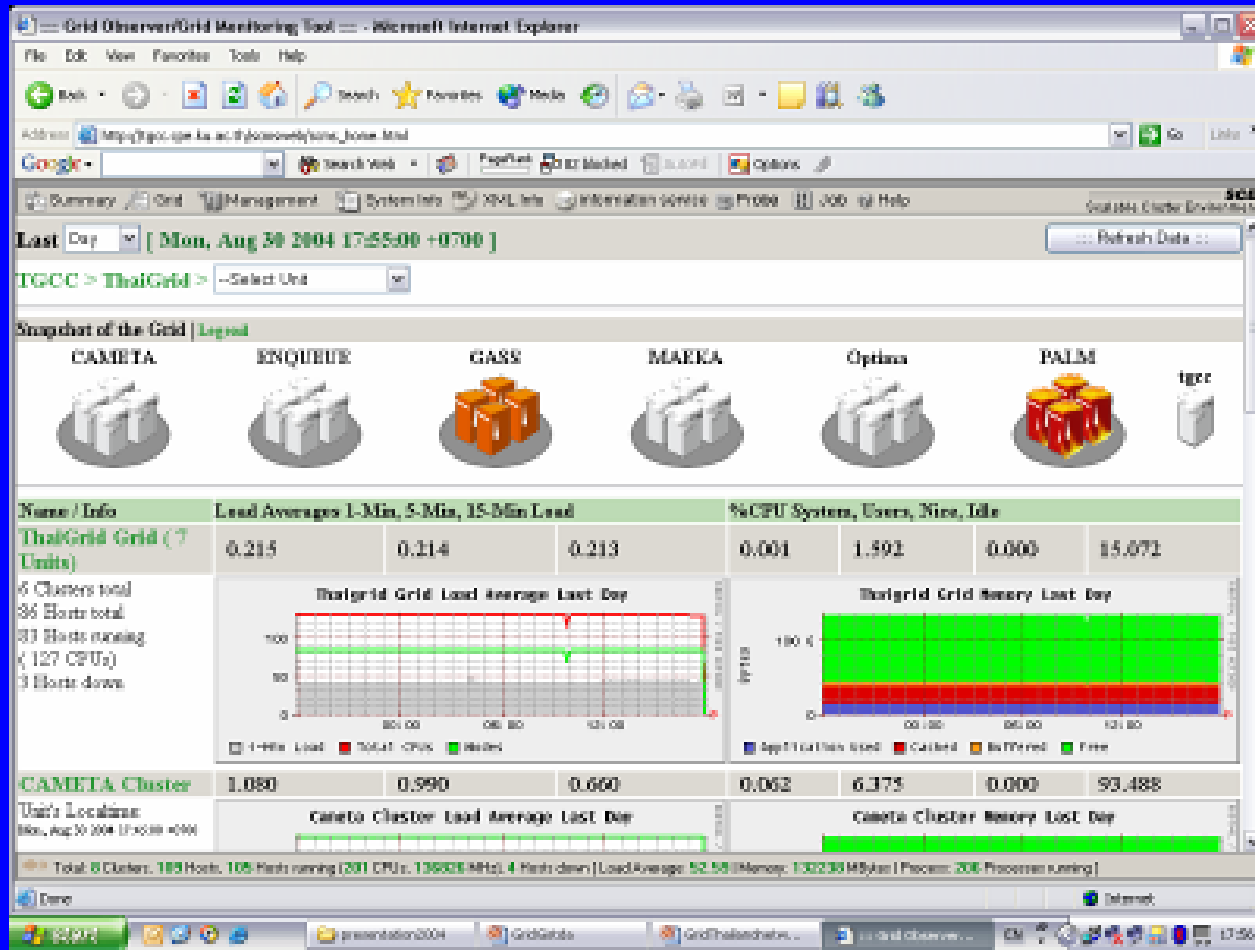
GISTDA

Grid Activities in Thailand

- Grid technology is
 - Being explored by university sector
 - Collaboration projects
 - ThaiGrid
 - ApGrid, APAN, PRAGMA
 - Still under experimental phase in Thailand

ThaiGrid

- A partnership project for grid computing started since December 2000
 - Kasetsart University
 - King Mongkut's Institute of Technology North Bangkok
 - Suranaree University of Technology
 - Asian Institute of Technology
 - Chulalongkorn University
 - Walailak University
 - Chiangmai University
 - KMUTT
 - NECTEC (National Electronics and Computing Technology)

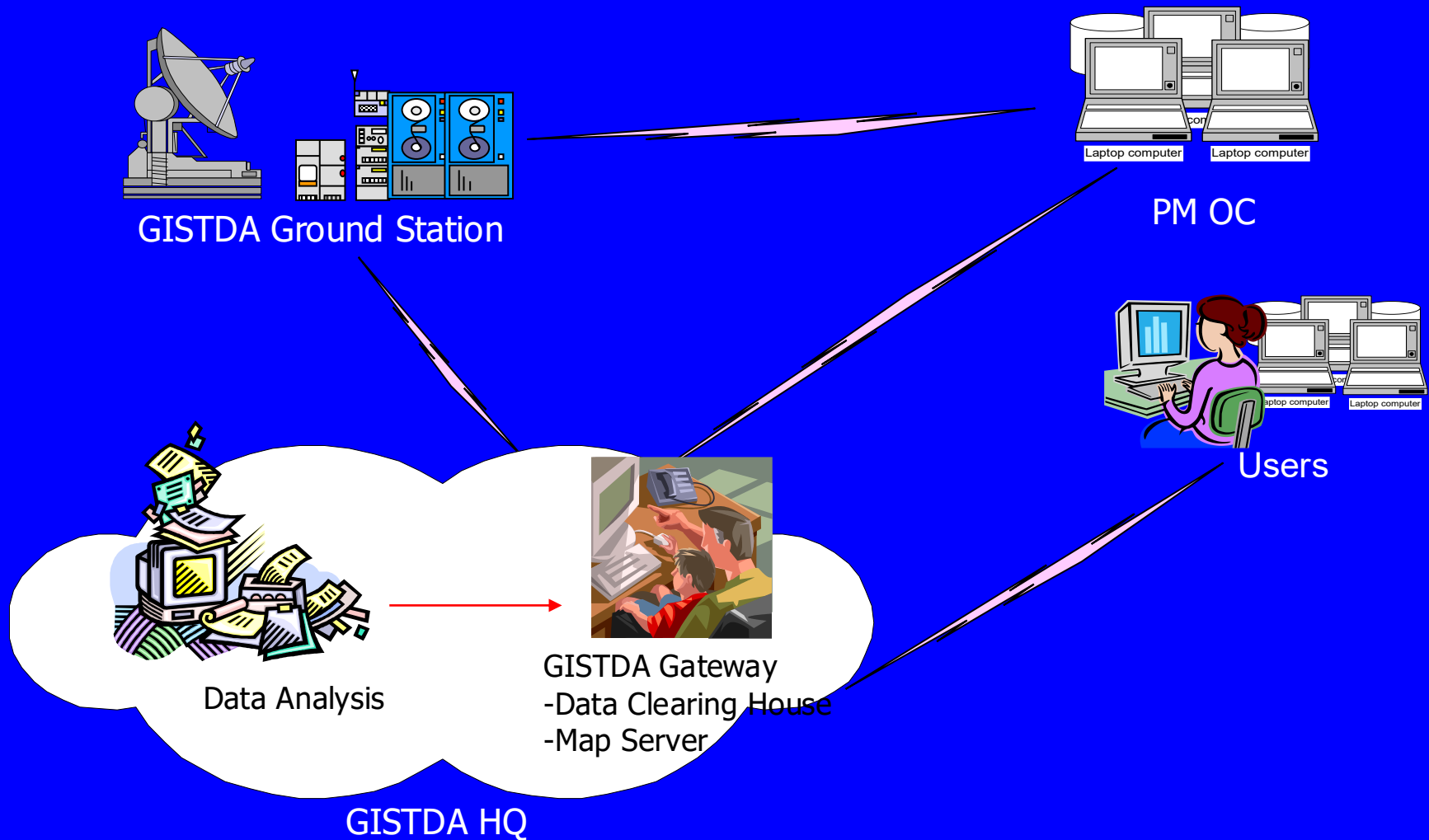


www.thaigrid.net

ThaiGrid Testbed

- 6 Universities collaboration
- 100+ processors

Pilot study: GISTDA Grid Computing



Challenges !!

- How to build a ubiquitous data sharing infrastructure among geographically distributed sites
- How to process, manage, store large data set
 - Transfer large satellite images between “workstation-server” and “server-server”
- How to build secure environment for organization critical data

Status

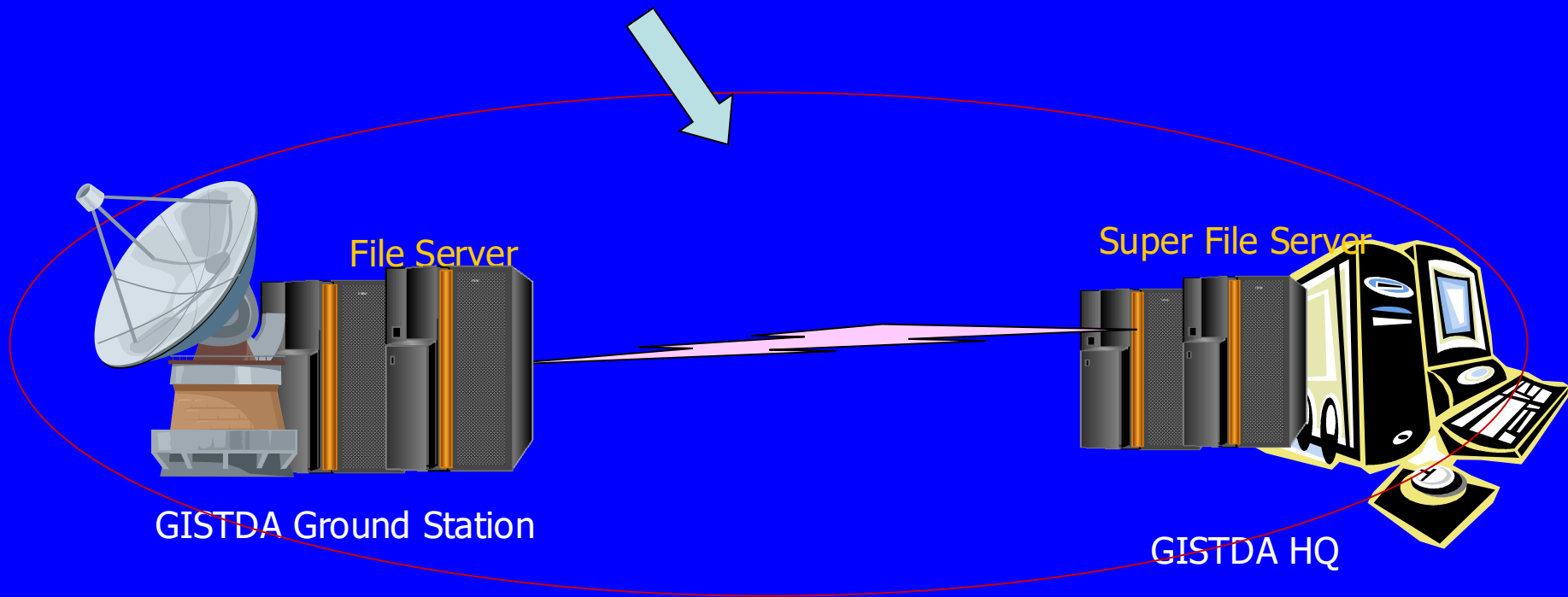
- Encouraged by WGISS/CEOS
- A pilot project is funded to explore the challenges/ problems involve –June 2004
- System will be implemented in couple month

Solution

- Using Grid and cluster computing
 - Using grid replica service to unify geographically distributed data.
 - Use cluster and parallel file systems technology to provide large and scalable file storage at economical cost
 - Deploy grid middleware (Globus 3.2) to build a secure infrastructure

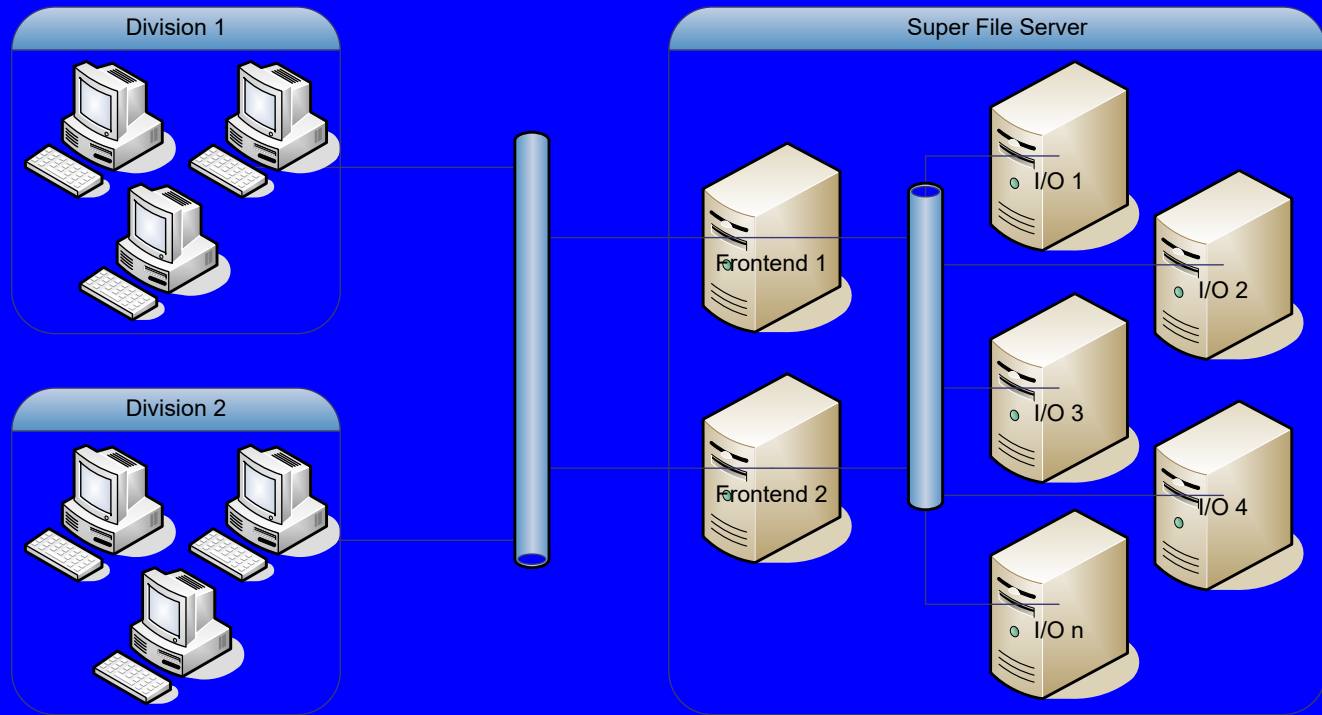
Data Grid & Replica Catalogue

Data Grid technology will be deployed to link 2 sites



“Using grid replica service to unify geographically distributed data”

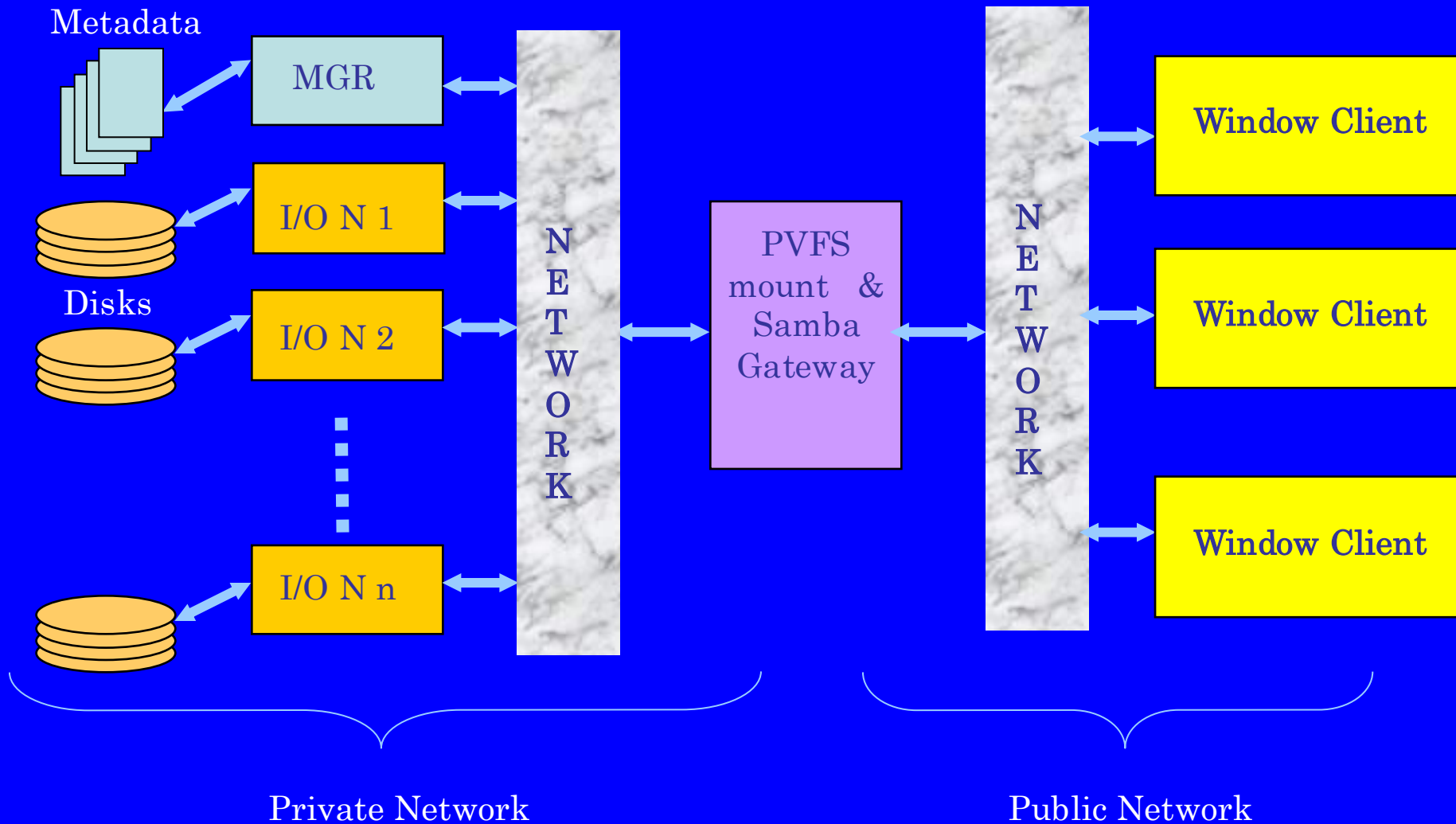
Configuration



“Use cluster and parallel file systems technology to provide large and scalable file storage at economical cost”

- Cluster front-end is configured to allow workstations to seamlessly get the data from back-end
- Multiple I/O nodes work collectively using parallel file system (PVFS)

Applies Storage Server <= PVFS + Samba



Expected Benefits

- Reduce the cost of building scalable storage by making use of open source technology
- Reduce access time for large file using powerful cluster file server
- Strong secure infrastructure using grid security
- Building computing power for further complex applications.

Conclusion

- Grid and cluster have been considered as potential technology to build a strong and secure IT infrastructure
- Wider deployment in the future using experiences learned